

SIH25170 Frontend Reference

Approved direction: Simple for everyone, powerful when needed.

Product Goal

Build a simple Earth Observation analysis application powered by GPT-OSS. A farmer or non-technical user should not need to understand AI, GIS, satellite sensors, bands, or remote sensing.

Main Flow

Select an area or upload EO imagery → confirm data/date → analyze → see image → see simple findings → ask a question → inspect visual evidence → optionally compare dates → optionally view technical details.

Default Screen

Prioritize Your Area, the main EO image/map, What We Found, What Changed, and Ask a Question. Avoid dashboards packed with charts.

Simple vs Expert

Simple Mode is the default and uses everyday language. Expert Mode is optional and contains sensor, source, resolution, coordinates, bands, confidence and processing details.

Visual Style

Light interface, white surfaces, soft borders, restrained green/blue/purple accents, readable typography and comfortable whitespace. Avoid dark futuristic dashboards, neon, glassmorphism, excessive gradients, glowing borders and decorative 3D elements.

AI Language

Use phrases such as 'What changed here?', 'Vegetation decreased', 'Buildings increased', and 'No major change found.' Avoid technical wording in the default UI.

Agent Rules

Reuse the design system. Do not redesign pages independently. One primary action per screen. Never invent data, statistics, confidence, imagery or AI results. Use clearly replaceable mock API data during development.

ML Boundary

Frontend consumes stable API responses. Keep API logic separate from UI. Handle queued, processing, completed and failed states, plus missing geometry/statistics.

Agent Reading Order

- 00_READ_ME_FIRST.md
- 01_FRONTEND_MASTER_SPEC.md
- 02_INFORMATION_ARCHITECTURE.md
- 03_UI_DESIGN_SYSTEM.md
- 04_PAGES_AND_COMPONENTS.md
- 05_SIMPLE_USER_FLOW.md
- 06_ANALYSIS_WORKSPACE_SPEC.md
- 07_FRONTEND_BACKEND_CONTRACT.md
- 08_AGENT_BUILD_RULES.md
- 09_VISUAL_REFERENCE.md
- 10_IMPLEMENTATION_CHECKLIST.md